

Raghav Shah

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Summary

Machine Learning Engineer with expertise in developing and deploying production-ready AI solutions on AWS. Skilled in prototyping and optimizing models for computer vision and high-dimensional embeddings using Python, PyTorch, TensorFlow, and SageMaker. Achieved a 40% reduction in inference time and built scalable pipelines that streamlined annotation workflows and cut operational costs. Passionate about translating research into practical healthcare applications and embedding ML capabilities into software platforms.

EDUCATION

Illinois Institute of Technology

Aug 2024 - May 2026

Master, Artificial Intelligence (GPA: 3.88/4.0)

- **Coursework:** Machine Learning, Advanced Artificial Intelligence, Time Series Analysis and Forecasting, Software Project Management, Data Preparation and Analysis, Big Data Technologies, Natural Language Processing

Chandigarh University

Aug 2020 - May 2024

Bachelor, Computer Science (GPA: 8.27/10)

WORK EXPERIENCE

CCC Intelligent Solutions | Data Science Engineer Intern

May 2025 - Present

- Optimized and deployed computer vision models using AWS SageMaker and Kubernetes, reducing inference time by 40% and aligning model delivery with production-ready practices by leveraging Python and ML frameworks. Migrated deployed models and services to AWS Graviton, creating cost-efficient pipelines that reduced infrastructure costs
- Engineered a custom data annotation platform to streamline segmentation tasks and visualize high-dimensional embeddings, boosting annotation speed by 50% and saving over \$20K annually in operational costs.
- Designed and deployed a scalable visualization tool for exploring 100K+ image embeddings, facilitating segmentation QA and improving model iteration cycles by 30% while enhancing debugging accuracy.
- Developed and maintained model inference pipelines for computer vision models using PyTorch and TensorFlow on AWS SageMaker, orchestrating deployment workflows via AWS Step Functions to automate car damage detection, tag prediction, and production releases, cutting manual deployment time by 50%.

Edifecs Technologies | Associate Data Engineer

Jan 2024 - Aug 2024

- Managed deployment of 15+ Health Insurance Enrollment Software suites and hotfixes on client premises and AWS infrastructure, enabling efficient processing of 1M+ EDI and 834 Enrollment files while ensuring HIPAA compliance in U.S. healthcare systems.
- Customized and optimized Health Insurance Enrollment Software for 10+ U.S. healthcare clients using AWS services (EC2, S3, Lambda, RDS, Step Functions) and containerized deployments with Docker, enhancing performance and scalability by reducing transaction processing time by 20% and improving reliability in enrollment and transaction management.

Celebel Technologies | Data Science Intern

Jun 2023 - Dec 2023

- Developed interactive dashboards using Matplotlib and Pandas for exploratory data analysis to inform model hypothesis verification and improve product quality by 20%.
- Applied deep learning techniques with PyTorch to enhance product design insights, reducing design iteration time by 15% and streamlining quality inspection with AI-driven solutions.

PROJECTS

MedCompare – AI-Driven Medication Data Evaluation Platform | Leap Of Faith Technologies

Apr 2025

- Built a benchmarking engine to evaluate AI-generated drug data across 400+ medications using fuzzy logic, semantic similarity scoring, and LLMs (OpenAI, Gemini, Llama, DeepSeek), achieving 87% overall accuracy and reducing validation time by 60% compared to manual review.
- Architected and deployed data pipelines to process 10,000+ JSON records, normalize outputs, and validate 10+ clinical fields (e.g., dosage, adverse reactions), enabling integration with FHIR-compliant systems for real-world clinical use.

Hybrid AgileGen | AI-Driven Software Planning

Nov 2024

- Combined Scrum and Waterfall methodologies to plan a web-based mobile app, creating 10+ use cases, 30+ requirements, and 5 workflows for Requirements, Design, Implementation, Testing, and Documentation.
- Utilized GenAI tools Autogen, LangChain, and LangGraph along with LLMs (OpenAI, Gemini) to generate 20+ outputs, boosting planning efficiency by 30% and revamping Scrum-based sprints.

SKILLS

- **Machine Learning & AI:** Scikit-learn, NumPy, Pandas, NLTK, LLMs, Langchain, OpenCV, TensorFlow, PyTorch
- **Programming Languages:** Python, C++, R, Java, C#, JavaScript, Kotlin, HTML, CSS, XML
- **Databases:** SQL, Firebase, SQLite, MongoDB, AWS DA
- **Frameworks & Libraries:** React, Node.js, Flask
- **Development & Tools:** Data Structures & Algorithms, SaaS, Git, GitHub, AWS DynamoDB, Amazon Bedrock AgentCore-, Amazon SageMaker, Amazon Bedrock, AWS Step Functions, AWS Lambda, Amazon EventBridge, Amazon EC2, ECR, Amazon S3, CI/CD, AWS, Docker
- **Additional Skills:** Statistics, Managing Ambiguity, Predictive Modeling, Cloud Technologies, Data Engineering, Curiosity

PUBLICATIONS

- Enhancing Machine Learning Model Using Explainable AI.. *Integrated LIME and SHAP techniques, highlighting top 5 features influencing individual predictions, and increased model transparency and trustworthiness, achieving 98% accuracy on Hotel Review Management Model. Authored and published technical paper on project in Springer LNNS, volume 796, Jan 2024, demonstrating model effectiveness and interpretability techniques.*

ACHIEVEMENTS/HONORS

- **Merit Sponsorship:** Received 40% merit sponsorship from IITC, 2024
- **Vice President, IEEE Student Club:** Vice president - IEEE Student club, Chandigarh University, 2023